MILITARY SPECIFICATION

TEST SET, RADAR, AN/UPM-136( )

1. SCOPE

1.1 This specification covers Radar Test Set AN/UPM-136. The test set is designed to check IFF (Identification Friend or Foe) transponders with certain associated equipment.

1.2 The test set is intended for use with Transponders resulting from the AIMS (Air Traffic Control Radar Beacon System/IFF/Mark XII Systems) Program, but shall also be capable of testing pre-AIMS equipments such as any of the Mark X SIF Transponders.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-17/30	- Cable, Radio Frequency, Coaxial, RG-62A/U (NWR-12)
MIL-C-17/84	- Cable, Radio Frequency, Coaxial, RG-223/U
MIL-S-901	- Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment and Systems, Requirements for
MIL-C-3432	- Cable and Wire, Electrical (Power and Control, Flexible and Extra Flexible, 300 and 600 Volts)
MIL-C-3767/12	- Connector, Plug, Electrical (Power, Three-wire, Polarized, Spring loaded Pivoted Grounding Type) Type UP131M
MIL-C-5541	- Chemicals Films and Chemical Film Materials for Aluminum and Aluminum Alloys
MIL-R-9673	- Radiation Limits, Microwave and X-radiation Generated by Ground Electronic Equipment (As Related to Personnel Safety)
MIL-Q-9858	- Quality Program Requirements
MIL-E-16400	- Electronic Equipment, Naval Ship and Shore, General Specification (Navy)
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-A-27434	- Adapter, Connector, Coaxial, Radio Frequency, Between Series, and Within Series, General Specification for
MIL-C-39012/1	- Connector, Coaxial, Radio Frequency, (Series N (Cabled)-Plug, Class li)
MIL-C-39012/6	- Connector, Coaxial, Radio Frequency, (Series C Plugs, Class li)
MIL-C-39012/12	- Connector, Coaxial, Radio Frequency, (Series C Receptacle, Flange Mounted, Rear Mounted, Class li)
MIL-C-39012/16	- Connector, Coaxial, Radio Frequency, (Series Bnc (Cabled)-Plug, Male, Class li)
MIL-C-39012/21	- Connector, Coaxial, Radio Frequency (Series Bnc (Uncabled)-Receptacles, Female, Jam Nut Mounted, Class li)
MIL-C-39012/22	- Connector, Coaxial, Radio Frequency, (Series Bnc (Uncabled)-Receptacles, Female, Flange Mounted, Class li)
MIL-C-39019	- Circuit Breakers, Magnetic, Low Power, Sealed, Trip Free, General Specification for

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	- Definition of and Basic Requirement for Enclosure for Electric and Electronic Equipment

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-470 - Maintainability Program Requirements (for Systems and Equipment)
MIL-STD-471 - Maintainability Demonstration
MIL-STD-681 - Identification Coding and Application of Hookup Wire and Lead Wire
MIL-STD-756 - Reliability Prediction
MIL-STD-781 - Reliability Tests Exponential Distribution
MIL-STD-785 - Reliability Program for Systems and Equipment Development and Production
MIL-STD-810 - Environmental Test Methods
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment, and Facilities
MS3102 - Connector, Receptacle, Electric, Box Mounting
MS3108 - Connector, Plug, Electrical, Angle 90 Degree
MS17321 - Meter, Time Totalizing Miniature, Digital 28 Volt DC

DRAWINGS

NAVSHIPS
REB-49224 - UG-60E/U Jack for Use with RG-8/U and RG-9/U Cables
REB-49227 - UG-59E/U Plug for Use with RG-8/U and RG-9/U Cables
REB-49228 - UG-61E/U Connector Receptacle for Use with RG-8/U and RG-9/U Cables
REC-49028 - Connector, Receptacle, Series BNC, UG-625B/U

HANDBOOKS

MILITARY
MIL-HDBK-217 - Reliability Stress and Failure Rate Data for Electronic Equipment

PUBLICATIONS

NAVSHIPS
0967-146-0010 - Dictionary of Standard Terminal Designations for Electronic Equipment

(Copies of specifications standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

US National Standards for Common System Component Characteristics for the IFF Mark X (SIF) Air Traffic Control Radar Beacon Systems SIF/ACTCRBS.

(Application for copies should be addressed to the Federal Aviation Administration, 800 Independence Avenue, SW, Washington, D.C., 20590.)

3. REQUIREMENTS

3.1 General requirements. The equipment shall conform to the requirements of MIL-E-16400, except as hereinafter specified.

3.2 First article sample. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.4).

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3.3 Descriptions. The descriptions of the Technical Standards for the Air Traffic Control Radar Beacon System (ATCRBS)/IFF/Mark XII Systems (AIMS) apply except as hereinafter specified.

3.3.1 Pulse shape. Pulse Shape parameters shall conform to figure 1 to aid in clarifying the descriptions. "Emax" is the maximum voltage amplitude of the pulse as measured from the zero axis:

(a) Points C and D on the pulse envelope are at the 90 percent Emax level.
(b) Points A and F on the pulse envelope are at the 10 percent Emax level.
(c) Points B and E on the pulse envelope are at the 50 percent Emax level.

3.3.2 Pulse rise time. The rise time of a pulse is the time required for the leading edge of the pulse to rise from 10 to 90 percent of the maximum pulse amplitude. T_R, as shown on figure 1, is the rise time and is the time required for the voltage to rise between Points A and C, except as hereinafter specified, pulse rise time shall be between 0.05 and 0.10 microseconds (μsec).

3.3.3 Pulse decay time. The decay time of a pulse is the time required for the trailing edge of the pulse to decay from 90 to 10 percent of the maximum pulse amplitude. T_D, as shown on figure 1, is the decay time required for the voltage to decay between Point D and F, except as hereinafter specified, pulse decay time shall be between 0.05 and 0.2.

3.3.4 Pulse top. The pulse top shall be essentially flat. The instantaneous amplitude of the pulse between the 90 percent Emax levels shall not fall below the 90 percent Emax level. On figure 1, the amplitude of any point between C and D shall not fall below the level of the line drawn through C and D.

3.3.5 Pulse duration. Pulse duration is the time between Points B and E, which are the 50 percent maximum voltage amplitude points on the leading and trailing edges. T_X, as shown on figure 1, is the duration time.

3.3.6 Delay or spacing. Delay or spacing is the time between 50 percent maximum voltage amplitude points on the leading edges of the pulses concerned, except as hereinafter specified.

3.3.7 Negative overshoot. The negative overshoot of a pulse during decay shall be less than 5 percent of Emax, the maximum pulse amplitude.

3.3.8 Normal pulse train. A pulse train shall consist of from 2 to 14 pulses and is delivered at the synchronizing repetition rate. The first and last pulses of any train, that is, the start and stop pulses are called the framing pulses. All other pulses located between the framing pulses at discrete spacings from the first framing pulses are called information pulses. Nominal pulse width shall be 0.45 μsec.

3.3.8.1 Mode 4. The Mode 4 pulse train shall consist of the pulse trains with the spacing specified in 3.8.2.2.5 and 3.8.2.2.6.

3.3.9 Repetition rate. Except as hereinafter specified, the requirements of this specification shall assume repetition rates of from 10 to 3,000 pulses, paired pulses, groups of pulses, or pulse trains per second.

3.4 Reliability. The test set shall have a specified mean-time-between-failure (θ_0) (MTBF) of 1500 hours, where θ_0 is defined by MIL-STD-781.

3.4.1 Reliability program. The supplier shall submit a reliability program plan in accordance with and in paragraph sequence with MIL-STD-785. The reliability program plan shall be submitted to the command or agency concerned for approval (see 6.3).

3.4.1.1 Reliability prediction. The supplier shall prepare reliability predictions in accordance with MIL-STD-756 and MIL-HDBK-217. The reliability predictions shall be submitted to the Command or Agency concerned for approval (see 6.3).

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3.4.1.2 Reliability demonstration. The reliability of the test set shall be demonstrated in accordance with the requirements of MIL-STD-781 and 4.5.

3.4.1.3 Internal temperature. During the reliability tests, the average internal ambient temperature within the equipment shall not exceed 25 degrees C beyond the external ambient under high line conditions.

3.4.2 Maintainability requirements. The test set shall have a mean corrective maintenance downtime (MCT) of 100 minutes and a maximum corrective maintenance downtime (MMAX CT) of 200 minutes at the 90th percentile.

3.4.2.1 Maintainability program plan. The supplier shall submit a maintainability program plan in accordance with and in paragraph sequence with MIL-STD-470. The program shall be a separate and distinct entity and shall be submitted to the Command or Agency concerned for approval (see 6.3).

3.4.2.2 Maintainability demonstration. Maintainability shall be demonstrated as specified in 4.6.

3.5 Environmental conditions.

3.5.1 The equipment shall conform to the temperature requirements of Class 2 except as hereinafter specified.

3.5.2 Temperature:

Operating: -28°C to +65°C
Nonoperating: -62°C to +75°C

3.5.2.1 The temperature of all operating panels shall be less than 43 degrees C when the ambient temperature in the space is 25 degrees C.

3.5.3 Altitude:

Operating: 0 to 10,000 feet above sea level
Nonoperating: 50,000 feet above sea level maximum

3.5.4 Vibration. The test set shall conform to the requirements of MIL-STD-167, Type I, as specified in MIL-E-16400.

3.5.5 Shock. The test set shall conform to the requirements for Class I, Type A, Grade A of MIL-S-901 except as hereinafter specified.

3.5.6 Inclination. The test set shall conform to the inclination requirement of MIL-E-16400.

3.5.7 Humidity. The test set shall be capable of operation at relative humidities ranging up to 95 percent for both continuous and intermittent periods, including conditions wherein condensation takes place in the form of water and frost.

3.5.8 Enclosure. The equipment shall be dripproof (15 degrees) in accordance with MIL-STD-108, as specified in MIL-E-16400.

3.6 General design.

3.6.1 Electrical design. The nominal supply voltage and frequency shall be 115 volts and both 60 and 400 Hertz (Hz).

3.6.1.1 Overload protection. Overload protection shall be provided. The type of protection and type of indicator shall be subject to the approval of the Command or Agency concerned.

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3.6.1.2 Input cable. The primary power input cable (see 3.8.8) shall connect to the equipment via a connector type 14S-3P in accordance with MS3102 or equivalent which shall be part of the equipment case.

3.6.1.3 Cables and connectors. All accessory cables and connectors required for complete and proper operation of the test set shall be provided by the supplier.

3.6.1.4 Input voltage. The equipment shall operate within the requirements of this specification from single phase supplies having an ac line voltage of 115 volts ±15 percent at any frequency from 45 to 420 Hz. The supply voltage may have a form factor between 1.03 and 1.25, and an amplitude factor between 1.16 and 1.68.

3.6.1.5 Power input. The connector specified in 3.6.1.1 shall be provided at the rear of the equipment, and shall mate with the cable specified in 3.7.8 for the input power.

3.6.1.6 Power consumption. The manufacturer shall employ all methods feasible in the process or manufacture, commensurate with good engineering practice and the requirements of this specification to keep the power consumption of the equipment to a minimum. Under no conditions shall the power consumption exceed 175 watts.

3.6.1.7 Transients. Provisions shall be made to prevent damage to the Cathode-ray tube (CRT) or other parts when the equipment is turned on or off, or when the equipment is subjected to transients as specified in MIL-E-16400.

3.6.1.8 Terminal designations. Standard terminal numbers shall be subjected to the approval of the command or agency concerned and may be in accordance with 0967-146-0010. Terminals which are standardized include the following:

(a) All terminals of this equipment used for connecting units of this equipment to units of other equipment
(b) All terminals for primary power, trigger input, trigger output, video input, and video output used in interconnecting units of this equipment

3.6.1.9 Power supplies. All alternating current (ac) and direct current (dc) voltages required by the test set shall be supplied by a common power supply. The power ON-OFF switch, indicator lamp, and protective device shall be located in the front of the case.

3.6.1.10 Power increase. All power transformers shall be capable of withstanding a 25 percent increase in nominal line voltage for at least 30 minutes. During the power increase, the transformers shall be loaded.

3.6.1.11 Receiver sensitivity. The equipment shall be so designed as to be capable of accurately measuring the receiver sensitivity of IFF equipments to a level of -80 dBm minimum.

3.6.1.12 R.F. power density and x-ray data. The supplier shall submit radio frequency (rf) power density and x-ray characteristics data for the test set in accordance with MIL-R-9673 (see 6.3).

3.6.1.13 Techniques. This equipment shall employ solid state and microelectronic techniques to the maximum extent practicable.

3.6.1.14 Colors. The standard colors to be used in chassis wiring shall be as specified in MIL-STD-681, as required by MIL-E-16400.

3.6.1.15 Color coding. Color coding of tube sockets is not required.

3.6.1.16 Electromagnetic interference characteristics. The test set shall conform to the applicable emission and susceptibility requirements of Class IC of MIL-STD-461. An interference control plan shall be provided in accordance with MIL-STD-461.

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3.6.1.17 IFF system compatibility. The equipment shall be compatible with the US National Standards for Common System Component Characteristics for the IFF Mark X (SIF)/Air Traffic Control Radar Beacon System SIF/ATCRBS.

3.6.1.18 Radiation protection. Protection and shielding shall be provided to protect personnel from the harmful effects of X-rays and other radiation. The method to be used to secure this protection shall be subject to the approval of the command or agency concerned. No protection is necessary if the CRT is used with less than 10,000 volts, and if solid state rectification is used in high voltage lines.

3.6.1.19 Test points. Test points and test features are required and shall be provided in accordance with MIL-STD-415, and shall be subject to approval of the command or agency concerned (see 6.3)

3.6.2 Mechanical design.

3.6.2.1 Size. The size and mounting dimension of the equipment shall not exceed 14 inches high by 14 inches wide by 24 inches deep.

3.6.2.2 Weight. The weight of the test set shall not exceed 70 pounds (uncrated).

3.6.2.3 Finish. Aluminum parts shall be chemically treated in accordance with MIL-C-5541.

3.6.2.4 Panel protection. Handles shall be provided on the front panel of the test set to protect the front panel controls and indicators from damage and for use in handling the unit.

3.6.2.5 Accessibility. All parts of the equipment shall be accessible for complete maintenance at the equipment's permanent installation. The location of all adjustments (electrical or mechanical) shall be such that maximum safety to the equipment and operating personnel is insured. To aid in access to test points and parts where the equipment is installed at a high level (that is, five feet above the floor level) a "lock and pivot" slide arrangement shall be used (see 3.8.1.2).

3.6.2.6 Rivets. Rivets shall not be used as a means of attaching potentially replaceable parts to any other part of the equipment.

3.6.2.7 Tools. Tools, other than an ordinary screwdriver, shall not be required to gain access to the several chassis for servicing.

3.6.2.8 Circuit cards. The design and use of all cards shall be subject to the approval of the Command or Agency concerned.

3.6.2.9 Ventilation. All apertures, louvers, and other openings shall be screened or attenuated where there is a possibility of radio frequency (rf) leakage.

3.6.2.10 Air circulation. Sufficient internal air circulation and heat conduction paths shall be provided to all parts of the equipment to prevent the failure of parts due to excessive heating. The blower motor (if required) shall be subject to the approval of the command or agency concerned.

3.6.2.11 Space heaters. Space heaters shall be employed, if required, in units in order to insure operation under the humidity conditions specified in MIL-E-16400.

3.6.2.12 Transformer marking. All power transformers shall be marked with winding and terminal designations stenciled on the case or printed on a plate attached to the case. The marking shall be easily read when the transformer is installed in the equipment.

3.6.2.13 Warning plates. In the event that the equipment becomes unbalanced when opened, warning plate(s) shall be provided.

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3.7 Controls, indicators, and panel layout.

3.7.1 Controls, indicators, and panel layouts. The design of operator and maintainer panel layouts and controls shall conform to the criteria of MIL-STD-1472 and 16400, except that color coding shall be in accordance with MIL-STD-1472. The submittal requirements of MIL-E-16400 shall include outline drawings of off-the-shelf equipments, views of hidden controls and displays, a functional description of the controls, and color coding.

3.7.2 Control knobs. In the interests of conserving panel space, and with approval of the command or agency concerned, dual concentric control knobs may be employed to control related functions.

3.7.3 Illumination. Illumination of front panel controls is not required.

3.7.4 Instruction plates. Instruction plates are not required.

3.8 Equipment composition. The Radar Test Set AN/UPM-136 shall consist of the following:

(a) Equipment case as specified in 3.7.1
(b) Test set as specified in 3.7.2
(c) Accessories and accessory box as specified in 3.7.5

3.8.1 Equipment case, CY-( )/UPM-136.

3.8.1.1 General. The case shall be constructed in accordance with MIL-E-16400 and shall be dripproof in accordance with MIL-STD-108. Carrying handles shall be employed where required; the design and placement of the handles shall be approved by the command or agency concerned (see 6.3).

3.8.1.2 Case, electrical equipment. The case shall accept the test set specified in 3.8.2. The slide arrangement employed in the test set (case/chassis) shall be a "lock and pivot" type which permits withdrawal of the equipment chassis to stops and then provides for rotation of the chassis to ±45 degrees and ±90 degrees. Safety interlock and bypass switches in accordance with the applicable portions of MIL-E-16400 and MIL-STD-454 are required.

3.8.1.3 Power input. The connector specified in 3.8.1.2 shall be provided at the rear of the equipment, and shall mate with the cable specified in 3.8.8 for the input power.

3.8.2 Test set, TS-( )/UPM-136. For the purpose of outlining the required performance parameters in this specification, the test set consists of three functional areas:

(a) Coder section (video generation)
(b) RF section (rf generation, modulation, demodulation, frequency, prf and power measurement)
(c) Oscilloscope section

3.8.2.1 Coder. The coder section of the test set shall generate IFF challenge pulse trains and video signals as specified herein and shall operate in synchronization with the "O" trigger specified in 3.7.2.5.1.

3.8.2.2 Mode select control. A control on the front panel shall perform the function of mode selection. It shall be a six position control identified as MODE SELECT with positions identified as 1, 2, 3/A, C, 4 (Word A), and 4 (Word B). The coder output for the selected mode shall be as specified in 3.7.2.2.1 through 3.7.2.2.6.

3.8.2.2.1 Challenge - Mode 1. Two pulses spaced 3.00 ±0.05 μsec apart. Pulse width shall be 0.8 ±0.05 μsec.

3.8.2.2.2 Challenge - Mode 2. Two pulses spaced 5.00 ±0.05 μsec apart. Pulse width shall be 0.8 ±0.05 μsec.

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3.8.2.2.3 Challenge - Mode 3/A. Two pulses spaced 8.00 ±0.05 μsec apart. Pulse width shall be 0.8 ±0.05 μsec.

3.8.2.2.4 Challenge - Mode C. Two pulses spaced 21.00 ±0.05 μsec apart. Pulse width shall be 0.8 ±0.05 μsec.

3.8.2.2.5 Challenge - Mode 4 (Word A). A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 13, 15, 17, 20, 24, 26, 30, 33, 35, 38, 41, 43, 45, 48, 50, 52, 54, 58, 61, 63, 66, 68, and 71 μsec with respect to the leading edge of the first pulse. No pulse of the pulse train shall vary more than ±0.07 μsec from its stated position. Pulse width shall be 0.5 ±0.05 μsec.

3.8.2.2.6 Challenge - Mode 4 (Word B). A pulse train shall be generated consisting of 28 pulses spaced 0, 2, 4, 6, 10, 12, 15, 17, 19, 22, 25, 28, 30, 32, 35, 37, 40, 44, 48, 51, 53, 55, 57, 60, 62, 64, 66, and 70 μsec with respect to the leading edge of the first pulse. No pulse of the pulse train shall vary more than ±0.07 μsec from its stated position. Pulse width shall be 0.5 ±0.05 μsec.

3.8.2.3 ISLS control pulse. Provisions shall be made for combining the ISLS pulse with the coder output for the Modes 1, 2, 3/A, C and 4 challenges (see 3.7.3.5.1). When selected, the ISLS pulse shall be triggered 2.0 ±0.05 μsec following the challenge P1 pulse for Modes 1, 2, 3/A and C providing the specific P2 functional characteristics. For Mode 4, the ISLS pulse shall occur 8.0 ±0.07 μsec after the first sync pulse. The pulse width shall be 0.8 ±0.05 μsec for Modes 1, 2, 3/A and C, and 0.5 ±0.05 μsec for Mode 4.

3.8.2.4 Coder output. The coder output shall be suitable for modulating the signal generator as specified in 3.8.3.4.

3.8.2.5 Triggers. The test set shall generate a zero trigger and a delay trigger. The unit will operate self-synchronous at a continuously variable rate over a linear range of 10 to at least 3,000 pulses per second (pps), for both the "O" trigger and delay trigger by means of a front panel control. The change in pulse repetition frequency (prf) shall be directly proportional to the amount of variation of the front panel control within ±10 percent.

3.8.2.5.1 "O" (Zero) delay trigger. The output of the internal synchronization trigger generator is defined as the "O" delay reference.

3.8.2.5.2 Delay trigger. Provisions shall be made for delaying the start of the oscilloscope sweep from less than 0.5 μsec to at least 300 μsec. The delay shall be continuously variable by means of a front panel control.

3.8.3 RF section. The functional area of the test set herein referred to as the rf section shall contain circuitry as required to generate crystal controlled rf signals (in conjunction with the coder), demodulate rf inputs, measure power and frequency, and count demodulated pulses/pulse trains.

3.8.3.1 Transmitter characteristics. The transmitter section of the test set shall generate rf signals as hereinafter specified.

3.8.3.2 Function select control. A control on the front panel shall select the method of operation of the transmitter section. It shall be a three position control identified as FUNCTION with positions identified as FREQUENCY, POWER, and SELF TEST. (Suitable abbreviations are permissible subject to approval of the Command or Agency concerned).

3.8.3.2.1 Power. The transmitter output frequency shall be crystal controlled and shall operate at a frequency of 1030 megahertz (MHz) ±0.01 percent. The signal generator shall be modulated as specified in 3.8.3.4. Power shall be measured as specified in 3.8.3.7.2 and the fidelity of the reply pulses observed as specified in 3.8.3.7.1.

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3.8.3.2.2 Frequency. A frequency measuring circuit shall be provided (as specified in 3.8.3.7.5) to check that the transponder frequency is within the limits of 1087 and 1093 MHz.

3.8.3.2.3 Self test. A portion of the signal generator output (including ISLS pulse when selected) shall be demodulated and made available at the demodulator front panel jack for observation on an oscilloscope and to the pulse counter (3.8.3.8). RF pulse fidelity shall be preserved as specified in 3.8.3.7.1.

3.8.3.3 Power output. The transmitter power output level at the test set interface jack shall be adjustable over the range of - 20 dBm to - 80 dBm.

3.8.3.3.1 Output attenuator. An attenuator shall be provided, capable of varying the test set rf output continuously betwen 20 and 80 dB below 1 milliwatt, when matched to a 50-ohm line. A front panel control identified by the word "Output Attenuator" shall be provided for the attenuator. It shall have an indicator calibrated in dB below 1 milliwatt under matched load conditions (50-ohm). The indicator shall have a range of at least 60 dB divisions. The differential attenuation between any two consecutive 10 dB divisions shall be 10 ±0.5 dB.

3.8.3.3.2 Output level accuracy. The rf output shall be within ±1.0 dB of the indicator setting over the entire range of frequencies, pulse modulation, and environmental conditions.

3.8.3.4 Modulation. The signal generator shall be capable of being satisfactorily pulse modulated with pulse trains as specified in 3.8.2.2.1 through 3.8.2.2.6. The modulator shall have an on-to-off ratio of 60 dB or better. The signal generator shall automatically be modulated by the challenge pulse train selected by the controls specified in the coder section of this specification. The challenge pulse train shall include the ISLS pulse when selected by the control of 3.8.3.5.1.

3.8.3.4.1 Delay. The delay between the modulator pulse applied to the signal generator and the signal generator output pulse shall not exceed 0.3 μsec.

3.8.3.4.2 Modulation fidelity. When modulated by the challenge pulse trains specified herein, the rf output signal shall have the following characteristics:

(a) Rise time 0.05 to 0.1 μsec
(b) Decay time 0.05 to 0.15 μsec
(c) The change in pulse width (modulator input versus output) shall be less than 0.025 μsec
(d) The spacing between the rf output pulses shall correspond to the modulation pulses within ±0.01 μsec

3.8.3.5 ISLS rf level control. The ISLS rf pulse level shall be continuously variable from a value at least 0 dB above to at least 12 dB below the output level of the signal generator specified in 3.8.3.3.1. A front panel control shall be provided to adjust the ISLS level. Calibration indices (and detents if required) or switch positions shall be provided to indicate the equal ("O") and -9 dB levels within ±0.5 dB.

3.8.3.5.1 Inhibit control switch. A three position switch shall be provided on the front panel and shall perform the following:

(a) OFF - All ISLS functions inoperative
(b) SLS IN - The rf ISLS pulse provided as specified herein
(c) DISPARITY (HOLD TO TEST) - This shall be a spring loaded position such that when engaged the pulse in the 30 μsec position of the Mode 4 maintenance word selected (see 3.8.2.2) shall be deleted

3.8.3.6 RF input/output connector. A front panel rf connector shall be provided to allow the test set to be connected to the transponder under test. The front panel jack shall be a type "C" female connector UG-568/U in accordance with MIL-C-39012/12 or equivalent.

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3.8.3.7 Demodulation. The demodulation section shall consist of pulse fidelity, power, and frequency measuring circuits. The demodulated video shall be provided on a front panel connector type UG-625B/U in accordance with MIL-C-39012/21.

3.8.3.7.1 Pulse fidelity. RF pulse shape fidelity shall be preserved by the demodulation circuitry and provided on the video output when power (3.8.3.2.1) or self test (3.8.3.2.1) operation is selected. With the highest power level input, and the video output terminated by the oscilloscope, the distortion of pulse rise time, fall time, and pulse width shall not exceed ±0.02 μsec or 5 percent, whichever is greater.

3.8.3.7.2 Pulse power. The demodulator circuitry video output shall represent the pulse power of the input signal when power (3.8.3.2.1) or self test (3.8.3.2.3) operation is selected. A peak pulse power input of 100 watts shall produce a video output of 1.0 volts minimum. When up to 1.5 kilowatts peak rf power is supplied, the video output shall be reducible by detector bias to 1.0 volts. For self test operation, the peak pulse power of the signal generator shall produce a video output of 1.0 volts minimum. It shall be possible to determine the power output of the signal generator within ±1 dB using the power measurement circuitry of the test set.

3.8.3.7.3 Power measurement. The demodulator circuitry shall also provide a detector bias output for processing and presentation on the meter specified in 3.8.3.8.3.

3.8.3.7.4 Measurement accuracy. Measurement by use of the meter specified in 3.8.3.8.3 shall permit determination of the peak power input over the range from 0.1 to 1.5 kilowatts, at any frequency from 1083 to 1093 MHz. The accuracy shall be ±1.0 dB. The meter shall be marked to indicate when the rf input is within the acceptance limit of above .30 kilowatt.

3.8.3.7.5 Frequency measurement. In FREQUENCY operation (3.8.3.2.2) the unit shall be provided with a frequency determining circuit to accurately provide a tuning indication for an input rf pulse. The tuning indication shall be in the form of a video pulse suitable for presentation on the oscilloscope. The pulse shall have a minimum amplitude of 1.0 volt of an input rf peak power level of 100 watts. When used in conjunction with the oscilloscope (see 3.8.4) the equipment shall be capable of measuring the frequency of rf signals from 1087 to 1093 to an accuracy of ±0.5 MHz. A dial shall indicate the frequency and be marked to show when it is within the acceptance limits of 1087.5 to 1092.5 MHz.

3.8.3.8 Pulse counter. The test set shall contain a pulse counter which shall count the repetition frequency of the demodulator output (including SIF reply code trains generated by a transponder under test) for both the POWER and SELF TEST modes of operation.

3.8.3.8.1 Pulse counter ranges. Pulse counter indication shall be on the meter specified in 3.8.3.8.3 and there shall be two ranges:

(a) 0 - 350 pps
(b) 0 - 3500 pps

3.8.3.8.2 Pulse counter accuracy. The accuracy of the pulse counter shall be within ±5 percent at full scale and ±10 percent at 1/10 full scale.

3.8.3.8.3 Meter. The test set shall be provided with a multiscale meter which serves both as the pulse counter indication and as the readout for the power measuring circuitry (see 3.8.3.7.2) as selected by a front panel control.

3.8.3.8.3.1 Scales for that portion of the meter which determine power readings shall be marked in kilowatts and shall be selected as necessary to meet the accuracy requirements of 3.8.3.7.4.

3.8.4 Oscilloscope section. The oscilloscope section shall be a partial dual trace oscilloscope. Dual display shall be obtained by alternating sweeps of video input and markers.

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3.8.4.1 Circuit elements. The oscilloscope functions shall be as provided by five functional areas:

(a) Sweep unit
(b) Marker unit
(c) Display unit
(d) Vertical unit
(e) Calibration

3.8.4.1.2 Sweep generator unit. The sweep generator unit shall perform the following functions:

3.8.4.1.3 Sweep duration. The following calibrated sweep durations shall be available with the controls located on the front panel of the unit when triggered at the repetition rates specified in 3.3.9. All sweeps shall be of sufficient amplitude to fill the mask opening. Sweep speed accuracy shall be ±5 percent.

0.1 μsec/cm	0.1 msec/cm
0.2 μsec/cm	0.2 msec/cm
0.5 μsec/cm	
1.0 μsec/cm	
2.0 μsec/cm	
5.0 μsec/cm	
10 μsec/cm	
20 μsec/cm	
50 μsec/cm	

3.8.4.1.4 Sweep direction. The horizontal sweep shall be from left to right when facing the CRT.

3.8.4.1.5 Sweep trigger and repetition. The horizontal sweep of the oscilloscope shall be triggered by the trigger specified in 3.8.2.5.1 at the specified repetition rates. The timing markers (3.8.4.2) are displayed on alternate sweeps.

3.8.4.1.6 Sweep linearity. The 10 percent portion of the physical length of the sweep, beginning 10 and 80 percent, respectively, from the start of the sweep, shall not differ in time duration from that portion of its physical length lying between 45 and 55 percent from the start of the sweep, by more than ±5 percent.

3.8.4.2 Marker selection. A three position (1 μsec/OFF/SIF) crystal marker switch shall be located on the front panel of the unit. Markers shall be provided with a spacing of 1.0 ±0.02 μsec or 20.3 ±0.02 μsec and 24.65 ±0.02 μsec from an initial marker. The initial marker shall be nominally timed to occur 3.0 ±0.1 μsec after the last SIF challenge pulse. The width of the 1.0 μsec markers shall not exceed 0.05 μsec and the width of the SIF markers shall not exceed 0.20 μsec.

3.8.4.2.1 Marker display. The output of the crystal marker generator shall be in a form suitable for amplitude modulation of the CRT trace, consisting of sharp indices or "pips" superimposed upon a dc level. The deflection shall be negative. Adjustment of the vertical centering of the marker sweep may be internal.

3.8.4.2.1.1 Marker amplitude. A front panel control shall permit variation of the marker level over a range of less than 1/2 to at least 2 cm.

3.8.4.2.2 Marker trigger. When selected, markers shall automatically be displayed on alternate sweeps.

3.8.4.2.3 Marker spacing accuracy. The markers shall have an accuracy of ±0.02 μsec.

3.8.4.2.4 Marker phasing. A front panel control shall permit horizontal displacement of the markers over a range of ±1 μsec. Jitter and countdown shall be permitted for the 1 μsec markers for phase changes greater than 0.9 μsec.

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3.8.4.3 Display unit.

3.8.4.3.1 CRT. A three inch diagonal flat faced CRT shall be employed using "A" type presentation. A scale shall be provided, located at the vertical center line graduated in fifths of a centimeter to measure the vertical displacement of the beam. The vertical scales shall extend two cm above and below the center of the tube face. A similar scale shall be provided for horizontal measurements, and shall be located approximately on the horizontal center line of the tube face. The horizontal scale shall extend three cm on each side of the vertical center. The tube shall be an Amperex D.7-78 or equivalent.

3.8.4.3.2 Focus. A control shall be located on the front panel for controlling focus of the trace on the CRT.

3.8.4.3.3 Intensity. A control knob shall be located on the front panel for controlling intensity of the beam. The control shall have sufficient range to vary the intensity of the beam from high to complete cut-off.

3.8.4.3.3.1 Astigmatism. An astigmatism control shall be provided on the front panel to adjust the astigmatism of the trace on the CRT screen.

3.8.4.4 Horizontal and vertical centering. Control knobs shall be located on the front panel for controlling the following functions:

(a) Horizontal centering. The horizontal control shall produce a range of control sufficient to center the trace on the CRT screen. This control shall move the start or finish of the trace to within 1.5 cm of the center of the screen.
(b) Vertical centering. The video channel vertical centering control shall produce a vertical displacement of the trace of at least ±2 cm from the center position, for the main video channel.

3.8.4.5 Vertical amplifier. The vertical amplifier shall have the following characteristics, selected by means of its selector switch on the front panel. (The gain of the vertical amplifier shall be continuously adjustable between the selected sensitivity positions by means of a convenient control.) Undershoot and overshoot shall be less than 5 percent of the pulse amplitude with a pulse rise time of 0.05 μsec or less:

Sensitivity	Rise time
0.05 volt per cm	Less than 0.03 μsec
0.1 volt per cm	Less than 0.03 μsec
0.2 volt per cm	Less than 0.03 μsec
0.5 volt per cm	Less than 0.03 μsec
1.0 volt per cm	Less than 0.03 μsec
2.0 volt per cm	Less than 0.03 μsec
5.0 volt per cm	Less than 0.03 μsec
10.0 volt per cm	Less than 0.03 μsec
20.0 volt per cm	Less than 0.03 μsec

3.8.4.5.1 Vertical linearity. With the gain of the vertical amplifier adjusted to obtain a 2 cm deflection with a voltage of 2 volts peak applied to the video input jack, and with the video sensitivity selector set for normal sensitivity, an input voltage of 1.0 volt shall cause a deflection of 1.0 ±0.05 cm, and an input voltage of 3.0 volts shall cause a deflection of 3.0 ±0.25 cm. Base line shift shall be directly proportional to the duty factor of the video signal in the ac coupling mode only. The check shall be made with a 1 μsec pulse of both positive and negative polarity.

3.8.4.6 Video input jack. The impedance looking into the video input jack shall be not less than that presented by a network consisting of 750,000 ohms in parallel with a capacitance of 45 picofarads (pF). Specified probe accuracy shall be obtained with any of the probes meeting the requirements specified herein, when connected to the video input jack. The equipment shall not be damaged by dc voltages of 500 volts applied to the panel connector.

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3.8.4.7 A switch on the front panel shall provide for an ac or a dc coupled display of the video input.

3.8.4.8 Polarity. Positive input pulse applied to the video input jack shall cause upward deflection of the spot on the oscilloscope screen. Negative input pulses shall cause downward deflections of the spot.

3.8.4.9 Rise time. The characteristics of the vertical amplifier shall be such that the rise time (in μsec) of a video pulse viewed on the CRT shall be not greater than that given by the following equation:

$$t_{out} = \sqrt{t^2_{in} + 0.0009}$$

Where t out = rise time of output pulse (μsec)

Where t in = rise time of input pulse (μsec)

3.8.4.10 Decay time. The characteristics of the vertical amplifier shall be such that the decay time (in μsec) of a video pulse viewed at the output of the amplifier shall be not greater than that given by the following equation.

$$t_{1\,out} = \sqrt{t_{1\,in} + 0.0009}$$

Where t_1 out = decay time of output pulse (μsec)

Where t_1 in = decay time of input pulse (μsec)

3.8.4.11 Frequency response. The frequency response of the vertical amplifier shall be essentially flat (within ±1.5 dB) from dc to 8 MHz. The roll-off characteristic shall be such that it will be less than 3 dB down at 12 MHz. The phase characteristics shall be adequate to insure that the slope of a 5000 μsec pulse will be increased by not more than 10 percent. The higher frequency response and phase characteristics shall be adequate to insure the rise and decay characteristics specified in 3.7.4.5, 3.7.4.9, and 3.7.4.10.

3.8.4.12 Video calibrator. The video calibrator shall supply pulses as specified herein for calibration purposes to a calibration output jack type UG-625B/U, REC 49028, on the front panel.

3.8.4.12.1 Output. The output of this unit shall be a 5 volts peak to peak calibration square wave at a prf of 1000 Hz ±20 percent. The rise and decay time of the calibration signal shall be equal to or less than 1.0 μsec with an external load of 500 K ohms and 40 pF capacity. During the decay time, the pulse shall have a negative overshoot of not more than 10 percent of the maximum positive amplitude. The ac output impedance of the video calibrator shall not exceed 2,100 ohms. The amplitudes of the output pulses shall be accurate to ±3 percent.

3.8.5 Accessories. Accessories shall be provided as hereinafter specified.

3.8.5.1 Accessory box. An accessory box capable of being mounted on top of the test set and having the capacity to hold the accessories specified herein, plus technical manuals if required (see 6.3).

3.8.5.2 Video test probes. Three video test probes shall be supplied. The characteristics of the video probes shall be as shown in Table I.

Table I. Video test probe characteristics.

Attenuation ratios	Input capacity	Dielectric	
		Volts DC	Volts AC (Peak to Peak)
1:1	50 pF, or less	600	600
10:1	10 pF, or less	600	600
100:1	5 pF, or less	1500	4200

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The general assembly of the probe shall consist of a length of shielded flexible cables (5.5 to 5 feet long) terminated by a connector of the BNC series and the probe body. Each probe shall be suitably marked for identification. Design of the probe shall be subject to approval of the command or agency concerned.

3.8.5.3 High power rf cable. One ten foot length of rf cable, type RG-223/U in accordance with MIL-C-17/84 or equivalent, having a UG-626B/U plug, in accordance with MIL-C-39012/6 or equivalent on one end and a UG-1185A/U in accordance with MIL-C-39012/1 or equivalent on the other end shall be provided. This cable shall be marked to indicate its attenuation.

3.8.5.4 Video jumper cable. One eight inch length of rf cable RG-62A/U in accordance with MIL-C-17/30 having a UG-260D/U plug in accordance with MIL-C-39012/16 or equivalent on each end shall be supplied.

3.8.5.5 Adapters. The following adapters shall be supplied:

Type	Applicable document	Quantity
UG-565A/U N to C	MIL-A-27434	1

3.8.5.6 Connectors. Unless otherwise specified herein, all external jacks for video pulses and trigger pulses shall be similar to type UG-625B/U, REC 49028, or UG-290C/U of MIL-C-39012/22. Associated plugs shall be similar to UG-260D/U of MIL-C-39012/16 or equivalent. Adapters shall be provided as specified in 3.8.5.5. Unless otherwise specified herein, all rf external jacks shall be similar to type UG-60F/U of REB 49224, UG-61E/U of REB 49228, UG-290C/U of MIL-C-39012/22, associated plugs shall be type UG-59E/U of REB 49227 or UG-88E/U of MIL-C-39012/16, or plugs and connectors suitable for RG-223/U cable in accordance with MIL-C-17/84.

3.8.5.7 Service test cable. A service test cable shall not be required provided that the internal power cable is of sufficient length to permit full extension of the chassis from the case and access to all test points, parts, or assemblies required for complete maintenance of the test set.

3.8.6 Input voltage. The equipment shall operate within the requirements of this specification from single phase supplies having an ac line voltage of 115 volts ±15 percent at any frequency from 45 to 420 Hz. The supply voltage may have a form factor between 1.03 and 1.25, and an amplitude factor between 1.16 and 1.68.

3.8.7 Elapsed time indicator. A dc elapsed time indicator, type MS17321-1 in accordance with MS17321, shall be incorporated in the test set to record cumulative operating time. The meter shall be operative whenever the main power switch of the test set is turned to ON position.

3.8.8 Primary power cable. This shall be an eight foot length of cable type CO-03MGF(3/18)0330 conforming to the requirements of MIL-C-3432, as specified in MIL-E-16400. The connector on the power input end of the cable shall be a type UP131M in accordance with MIL-C-3767/12. The equipment chassis shall be electrically connected to the ground pin on the connector plug. The connector on the chassis entrance end of the cable shall be a plug (90 degrees), 14S-3P in accordance with MS3108.

3.8.9 Circuit breaker. Circuit breakers shall conform to MIL-C-39019 and shall be used for the primary input power circuitry.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality assurance program. The supplier shall provide and maintain a written quality program acceptable to the Government for supplies and services covered by this specification. The quality program shall be in accordance with MIL-Q-9858 (see 6.3).

4.1.2 Quality assurance plan. A quality assurance plan, applicable to the equipments covered by the specification, shall be prepared for Government approval (see 6.3).

4.1.3 Government verification. All quality assurance operations performed by the supplier will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all inspection requirements of this specification. Failure of the supplier to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.4 Test procedures. The supplier shall prepare and submit for Government approval test procedures covering first article and quality conformance inspections (see 6.3).

4.1.5 Quality assurance terms and definitions. Quality assurance terms, used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)
    (1) Production inspection (see 4.4.1)
    (2) Production control inspection (see 4.4.2)
    (3) Environmental inspection (see 4.4.3)
    (4) Reliability testing (see 4.5)
(c) Maintainability demonstration (see 4.6)
(d) Inspection of preparation for delivery (see 4.8)

4.3 First article inspection. Unless otherwise specified (see 6.2) one unit shall be required for first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the inspections specified in table I.

TABLE I. First article inspection

Examination or test	Requirement paragraph	Test method paragraph
Production inspection	3.1	4.4.1
Production control inspection	3.1	4.4.2
Environmental inspection	3.6	4.4.3

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Table I. First article inspection (Continued)

Examination or test	Requirement paragraph	Test method paragraph
Salt spray test	3.1	4.7.1
Inclination test	3.5.6	4.7.1
Enclosure test	3.5.8	4.7.7

4.3.1 First article inspection report. The supplier shall prepare and submit for approval of the Command or Agency concerned a first article inspection report covering the results of the first article inspection (see 6.3).

4.4 Quality conformance inspection.

4.4.1 Production inspection. Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests and examinations approved in accordance with 4.1.4 and shall include the requirements of table II.

TABLE II. Production inspection

Examination or test	Requirement paragraph	Test method paragraph
Surface examination	3.1, 3.6.2.3, 3.6.2.4	4.7.2
Operating test	3.1	4.7.3

4.4.2 Production control inspection. Production control inspection, including sampling, shall conform to table III and to the inspection procedures of MIL-STD-105 using the special inspection levels. The inspection level shall be S-3 for normal, tightened, and reduced inspection. Production control inspection shall be performed on equipments that have passed production inspection. Production control inspection shall be approved in accordance with 4.1.4. The equipment shall satisfactorily meet the requirements of production control inspection prior to release for shipment.

TABLE III. Production control inspection

Examination or test	Requirement paragraph	Test method paragraph	AQL (percent defective)
Weights and dimensions	3.6.2.1, 3.6.2.2	4.7.1	6.5
Supply line voltage & frequency tests	3.6.1	4.7.1	6.5
Heat test	3.6.2.10	4.7.1	6.5
Power test	3.8.3.7.2, 3.8.3.7.3	4.7.1	6.5
Controls and control circuits	3.7	4.7.1	6.5

4.4.2.1 Rejected lots. If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

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4.4.2.2 Reinspection of conforming production control test units. Unless otherwise specified, sample units which have been subjected to and passed production control inspections may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.4.3 Environmental inspection. Environmental inspection shall be as listed in table IV and shall be performed on units which have been subjected to and passed production inspection. Environmental tests shall be approved in accordance with 4.1.4.

TABLE IV. Environmental inspections

Examination or test	Requirement paragraph	Test method paragraph
Temperature tests	3.5.2	4.7.5
Humidity test	3.5.7	4.7.1
Shock test	3.5.5	4.7.6
Vibration test	3.5.4	4.7.1
Altitude test	3.5.3	4.7.4
Electromagnetic interference emission, susceptability, and equipment and systems self-compatibility	3.6.1.16	4.7.1
Noise test	3.1	4.7.1

4.4.3.1 Environmental inspection. Approval to ship may be withheld at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action.

4.4.3.2 Sampling for environmental inspection. One equipment shall be selected from each successive 100 equipments produced or fraction thereof. The first sample equipment shall be selected from the first month's production.

4.4.3.3 Nonconforming environmental sample units. If a sample unit fails the inspection specified in 4.4.3, the supplier shall immediately investigate the cause of failure and shall report to the Government Quality Assurance Representative the results thereof and details of the corrective action taken to correct units of product which were manufactured with the same conditions, materials, processes, and so forth. If the Quality Assurance Representative does not consider that the corrective action will enable the product to meet specified requirements, or if the supplier cannot determine the cause of failure, the matter shall be referred to the contracting officer (see 4.4.3.1).

4.4.3.4 Reinspection of conforming environmental sample units. Unless otherwise specified in the contract, sample units which have been subjected to and passed environmental inspection may be accepted on the contract provided they are resubjected to and pass production inspection after repair of all damage.

4.5 Reliability testing.

4.5.1 Reliability demonstration. Reliability demonstration shall be performed in accordance with MIL-STD-781.

4.5.1.1 Reliability demonstration test plans. The supplier shall prepare and submit to the Command or Agency concerned a reliability demonstration test plan which conforms to the detailed requirements of MIL-STD-781, and the requirements of this specification. The reliability test plans shall be submitted to the command or agency concerned for approval (see 6.3).

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4.5.1.2 Reliability qualification tests. Qualification testing shall be performed on at least two production units. Testing shall be performed at test level C with test plan V. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan V.

4.5.1.3 Reliability sampling tests. Reliability sampling tests shall be conducted throughout production to ensure continued production of acceptable equipment. Sampling testing shall be at test level C, with test plan VI of MIL-STD-781. At least three units shall be randomly selected from each six month's production for reliability sampling tests. Testing shall proceed to a decision in accordance with the criteria specified in MIL-STD-781 for test plan VI.

4.5.1.4 Failure. Failure shall be defined as any malfunction or parameter deviation that prevents the equipment from performing within the operational requirements set forth in this specification.

4.5.1.5 Rejected equipment. When equipment has been rejected as a result of the reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test. In the event that the cause of failure is equipment manufacturer design or part manufacturer designs, corrective action shall be performed on all equipment delivered under the contract.

4.5.1.6 Disposition of tested equipment. Upon successful completion of reliability testing, tested equipment shall be returned to the production area for reinspection and reconditioning as required and shall be subjected to the production tests of 4.4 to assure that it conforms to the requirements of this specification. After reconditioning, reinspection, and retesting, the equipment may be delivered as part of the contract quantity provided it meets all terms and conditions of the contract for acceptance.

4.5.1.7 Reliability reports. Reports, providing a complete accounting of progress on each element defined by the reliability program plan, results achieved, and status of actions to resolve problems and to correct weak links, shall be submitted, as specified in the contract or order, to the Command or Agency concerned (see 6.3).

4.6 Maintainability demonstration. A maintainability demonstration test shall be conducted in accordance with MIL-STD-471, Test Method 1, Plans A and B, to verify that the test sets comply with the requirements specified herein.

4.7 Inspection procedures.

4.7.1 Tests and examinations. Tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400, except when otherwise stated.

4.7.2 Surface examination. Each equipment shall be visually and mechanically inspected for compliance with the workmanship criteria of MIL-E-16400 and as defined in MIL-STD-454, requirement 9, to determine that materials, parts, finish, construction, marking, mechanical safety, assembly and fit, and workmanship requirements of this specification have been met.

4.7.3 Operating test. The equipment shall be completely aligned and calibrated and subjected to an operating test for 24 hours to insure qualitatively the proper functioning of the equipment, including all operating controls and conformance with safety requirements.

4.7.3.1 All power transformers shall be tested a minimum of 30 minutes under the following conditions:

(a) A 25 percent increase in nominal line voltage
(b) Transformer shall be loaded with nominal load

4.7.4 Altitude test. Altitude tests shall be performed on the complete equipment in accordance with Method 500, Procedure I, of MIL-STD-810 to show compliance with 3.5.3.

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4.7.5 Temperature tests. Nonoperating and operating condition temperature tests shall be conducted on the complete equipment in accordance with MIL-E-16400 to show compliance with 3.5.2. During the operating condition temperature tests the requirements of 3.5.2.1 shall be verified.

4.7.6 Shock. The test set shall meet the shock requirements of Grade A Deck Mounted, Class 1, Lightweight and Type A of MIL-S-901. The test set shall not be turned on during the test but shall perform all its functions within the limits specified in this specification at the completion of the test. The method of mounting the equipments for testing shall be subject to the approval of the Command or Agency concerned.

4.7.6.1 The CRT must be installed during the one-foot hammer drop and must operate within the limits of this specification afterwards. Distortion of the sweep is permissible after the three and five-foot drops or the CRT may be removed for these blows.

4.7.7 Enclosure test. Drip tests shall be performed on the complete equipment in accordance with MIL-E-16400 to show compliance with 3.5.8.

4.8 Inspection of preparation for delivery. Sample packages and packs and the inspection of the preservation and packaging, packing and marking for shipment and storage shall be conducted to insure compliance with the requirements of Section 5.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation and Packaging, Packing, and Marking. The equipment, accessories, technical publications and repair parts shall be preserved and packaged level A or C, packed level A, B, or C, and marked as specified in the contract or order in accordance with MIl -E-17555.

6. NOTES

6.1 Intended use. The equipment described in this specification is intended for Navy shipboard and shore use.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Number of first article samples to be submitted if other than specified (see 4.3)
(c) Marking, levels of preservation and packaging and packing required (see 5.1)

6.3 Contract data requirements. Data generated by this document are not deliverable unless specified on the Contract Data Requirements List (DD Form 1423) or the contract schedule. The data required by this specification include, but are not restricted to the following:

(a) Reliability program plan (see 3.4.1)
(b) Reliability predictions (see 3.4.1.1)
(c) Maintainability program plan (see 3.4.2.1)
(d) Technical manuals including operating instructions if required by the contract or order
(e) RF power density and X ray characteristics data (see 3.6.1.12)
(f) Quality assurance program (see 4.1.1)
(g) Quality assurance plan (see 4.1.2)
(h) Test procedures (see 4.1.4)
(i) First article inspection report (see 4.3.1)
(j) Reliability demonstration test plans (see 4.5.1.1)
(k) Reliability reports (4.5.1.7)

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6.4 First article. Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 6625-N479)

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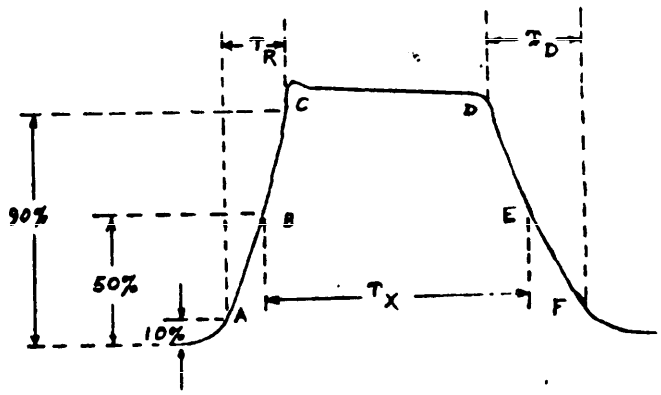


FIGURE 1. Pulse shape characteristics.

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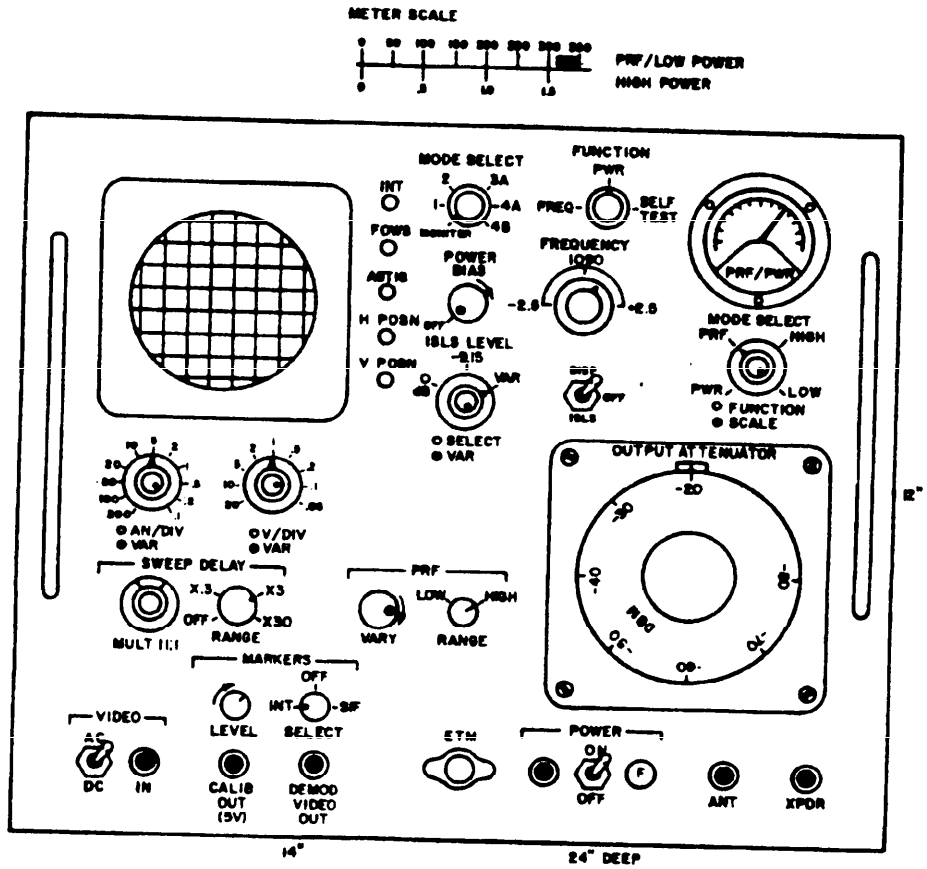


FIGURE 2. Test set, radar, AN/UPM-136( ).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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